

★ 1 Fundamental lung-ventilator coupling equation (core research equation)

The Equation of Motion of the Respiratory System

$$P_{aw}(t) + P_{mus}(t) = R_{RS} \dot{V}(t) + E_{RS} V(t) + P_{EEP}$$

This equation represents **force balance between ventilator pressure and lung mechanics** and is the primary mathematical description of ventilator–lung interaction.

👉 It states airway pressure must overcome:

- resistive load $R_{RS} \dot{V}$
- elastic load $E_{RS} V$
- end-expiratory pressure

This equation is explicitly described in ventilator mechanics literature.

✅ Direct link

👉 <https://academic.oup.com/ejhacc/article/14/8/494/8195106>

👉 <https://www.sciencedirect.com/science/article/pii/S0883944124003897>

★ 2 Differential form of equation of motion (research differential model)

Research literature states pressure, volume and flow are **continuous functions of time**, therefore the equation is treated as a differential equation.

$$P_{aw}(t) = P_{RES}(t) + P_{EL}(t) + P_{INERT}(t)$$

where

$$P_{RES} = R \dot{V} \quad P_{EL} = \frac{V}{C} \quad P_{INERT} = I \ddot{V}$$

This decomposition is explicitly described in respiratory mechanics derivation papers.

✅ Direct link



https://www.researchgate.net/publication/351245868_APPLICATION_OF_MODIFIED_DERIVED_EQUATIONS_OF_MOTION_OF_RESPIRATORY_MECHANICS

★ 3 Exponential lung filling equation (mechanical ventilation dynamics)

Ventilator literature shows lung filling follows:

$$V_t = V_0(1 - e^{-kt})$$

This exponential dynamic is reported in mechanical ventilation modelling research.

✓ Direct link

👉 <https://pmc.ncbi.nlm.nih.gov/articles/PMC10034927/>

★ 4 Compliance equation (peer-reviewed definition)

$$C = \frac{\Delta V}{\Delta P}$$

Compliance definition appears in respiratory mechanics studies describing lung elastic behavior.

★ 5 Resistance equation (peer-reviewed definition)

$$R = \frac{\Delta P}{\dot{V}}$$

This describes resistive pressure–flow relationship in ventilated lungs.

★ 6 Multi-compartment lung equation (heterogeneous lung research)

Research models describe multi-unit lung behavior as

$$P_{aw} = \sum_{i=1}^n (R_i \dot{V}_i + E_i V_i) + P_{EEP}$$

Used in ARDS and heterogeneous lung modeling literature.

✓ **Direct link**

👉 <https://arxiv.org/abs/2408.14607>

★ **7 Nonlinear lung model used in ventilator research**

Recent research introduces nonlinear pressure-volume relation:

$$P = aV^2 + bV + c$$

used to capture lung hysteresis and disease behavior.

✓ **Direct link**

👉 <https://arxiv.org/abs/2402.18709>

★ **8 Patient-ventilator interaction statement (research evidence)**

Peer-reviewed reviews state ventilator graphics and interaction are governed by the equation of motion describing patient-ventilator mechanics.

✓ **Direct link**

👉 <https://pmc.ncbi.nlm.nih.gov/articles/PMC8360707/>

★ **Important insight (very important for your project)**

All research digital twins of ventilated lungs ultimately use:

$$P_{aw} = R\dot{V} + EV + I\ddot{V} + PEEP$$

with possible extensions:

- nonlinear compliance
- multi-compartment lungs
- adaptive parameters
- gas exchange models

This is the **canonical research model**.

★ 1 Core lung–ventilator coupling (Equation of Motion)

The fundamental equation governing ventilator–lung interaction:

$$P_{aw}(t) = R_{RS} \dot{V}(t) + E_{RS} V(t) + EEAP$$

where

- P_{aw} airway pressure
- R_{RS} resistance
- E_{RS} elastance
- V volume
- $\dot{V}$ flow
- $EEAP \approx PEEP$

This equation is the basic respiratory mechanics model used to describe patient-ventilator interaction.

👉 Direct paper links

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC8360707/>
- <https://www.sciencedirect.com/science/article/pii/S0883944124003897>

★ 2 Extended equation of motion (including inertance)

$$P_{aw}(t) = R \dot{V}(t) + \frac{V(t)}{C} + I \ddot{V}(t) + PEEP$$

Pressure is decomposed into resistive, elastic and inertial components in respiratory mechanics studies.

👉 Direct link

- <https://www.arcjournals.org/pdfs/ijcclm/v7-i1/3.pdf>
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★ 3 Compliance definition

$$C = \frac{\Delta V}{\Delta P}$$

Respiratory compliance measures lung elastic behavior.

★ 4 Resistance definition

$$R = \frac{\Delta P}{\dot{V}}$$

Defines airway pressure-flow relationship in ventilated lungs.

★ 5 Time constant of lung mechanics

$$\tau = R C$$

Time constant describes speed of lung filling and emptying.

👉 Link

- <https://www.researchgate.net/publication/355033909>
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★ 6 Exponential lung filling model

$$V_t = V_0(1 - e^{-kt})$$

Ventilation dynamics commonly follow exponential filling behaviour.

👉 Link

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC10034927/>
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★ **7 Alternative exponential form**

$$V_t = V_i e^{-t/\tau}$$

Used in respiratory mechanics to model volume change with time constant.

👉 Link

- <https://journals.sagepub.com/doi/10.4187/respcare.03410>
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★ **8 Multi-compartment lung model**

$$P_{aw} = \sum_{i=1}^n (R_i \dot{V}_i + E_i V_i) + PEEP$$

Used in heterogeneous lung modeling and ARDS research.

👉 Link

- <https://arxiv.org/abs/2408.14607>
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★ **9 Nonlinear pressure–volume relation**

$$P = aV^2 + bV + c$$

Quadratic pressure-volume modeling captures nonlinear lung behaviour.

👉 Link

- <https://arxiv.org/abs/2402.18709>
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★ **10 Rapid shallow breathing index**

$$RSBI = \frac{f}{V_T}$$

Used clinically to assess ventilator weaning.

★ **1 1 Dead-space (Bohr equation)**

$$\frac{V_d}{V_t} = \frac{P_{ACO2} - P_{ECO2}}{P_{ACO2}}$$

Describes physiological dead space in lungs.

★ **1 2 Alveolar gas equation**

$$p_A O_2 = F_I O_2 (P_{ATM} - p_{H_2O}) - \frac{p_a C O_2}{RER}$$

Used to model oxygen exchange during ventilation.

★ **1 3 Ventilation rate equation**

$$\dot{V}_E = V_T \times f$$

Defines total ventilation per minute.